

ECEN 454/714 – Lab1: Introduction to Cadence Schematic Design & Simulation

We will now begin the design by implementing the logic design of the 8-bit Pipelined adder. The following sections introduce you to the procedures to use Cadence for schematic capture and simulation which you will use to implement the required logic design.

1. Introduction to Cadence tool

- If you see “***x11fonts: not found”, disregard it. You can also see two pop-up windows telling you what’s new about this version of Cadence. Click ok. At the same time, a small rectangular window will pop up. This is called Command Interpreter Window (CIW). This is where virtuoso will display all the messages about your commands. Do not close this window (except when quitting!) and always keep this window visible. CIW displays all the error and warning messages, so please look into this whenever possible. A window called Library Manager will also pop up.

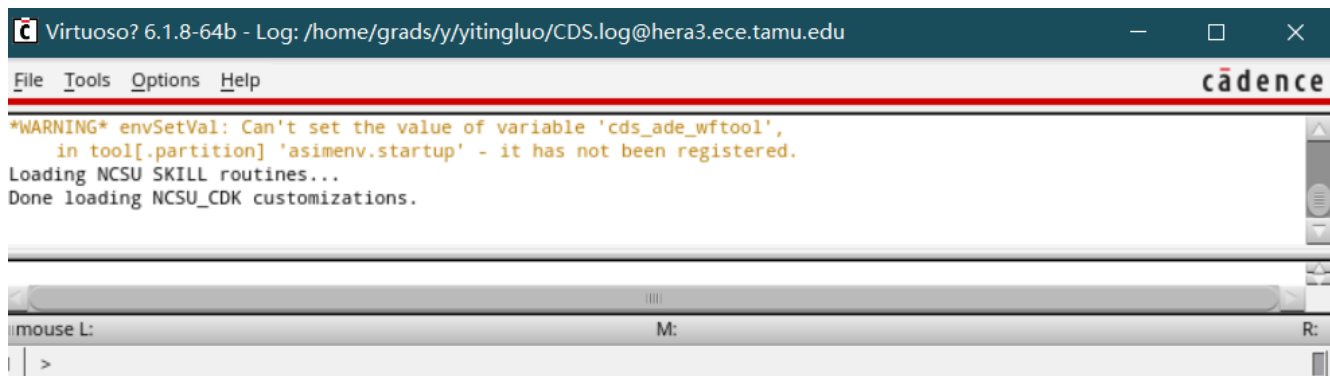


Figure 1. CIW Window

- The heading of the CIW is Virtuoso - Log: /homes/xxxx/CDS.log and has four menus File, Tools, Options and Help. The lower left corner has the virtuoso prompt which displays ">", when it awaits next action.
- Virtuoso uses Library Manager for browsing through all the libraries. The Library Manager has several libraries including any designs created by you. Several other predesigned libraries are also required before your own design. This is why file “cds.lib” was copied to your directory when you start cadence, make sure it exists under your current running directory.
- Make sure you’ve downloaded gdpk045_v_5_0 and unzip it. In the Library Manager window, select Edit > Library Path, and you will see the Library Path Editor window pops up.
- In the window, select File > Open to find and select the “cds.lib” file just saved in your cadence directory, click it, and press OK. You can simply ignore warning messages about unlocked file. If the “cds.lib” file is well launched, you will see the Library Path Editor window as below. Type gdpk in the column on the left, select the library path by right clicking and selecting “Add library” on the column on the right. In the Add Library window, find the gdpk045_v_5_0 folder, and select gdpk045. Press “OK”. Similarly, add assura to the library list as well.

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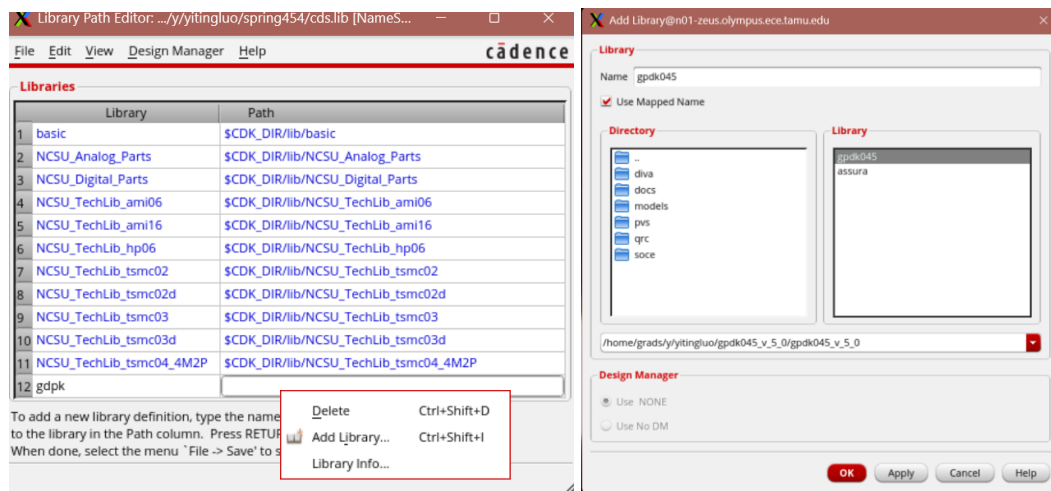


Fig 2. Library Management

The library list would look like this, choose File>Svae.

12	gpdk045	/home/grads/y/yitingluo/gpdk045_v_5_0/gpdk045_v_5_0/gpdk045
13	assura	/home/grads/y/yitingluo/gpdk045_v_5_0/gpdk045_v_5_0/assura

- At the Library Manager window, click File > New > Library and type in name “Design” as shown below. Click OK.

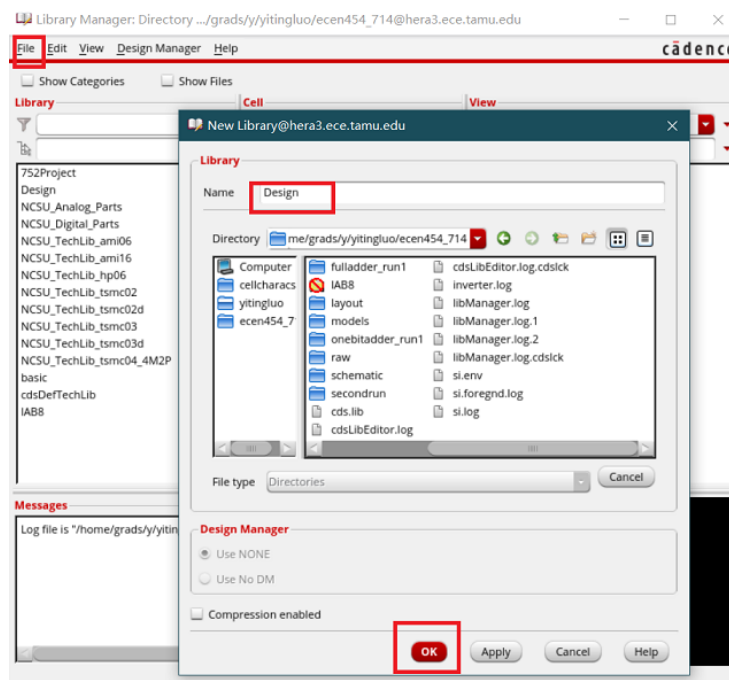


Figure 3. New library created.

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- Next, you will find a menu for technology files. Choose “Attach to an existing technology library” and click OK. Then choose “gdpk045” as shown below. It means from now on when you design a device, you will do it according to the rules related to this technology.



Figure 4. Attach gdpk045 library.

2. Schematic of the Full-adder Circuit.

The full adder is the basic building block of the 4-bit adder block that we need to design which we will eventually use to create the 8-bit Pipelined adder. The method is to create the design. (Schematic in this case) of the lowest building block (fulladder) and create a “symbol” of this design which can be reused in some other design (we would be using it in a higher-level design to create a 4-bit adder)

- In this lab we will design a one-bit adder. Let’s create a new cell called "fulladder". For this click on the “Design” library in the Library Manager window and choose File New > Cell view. Type cell name and view name.

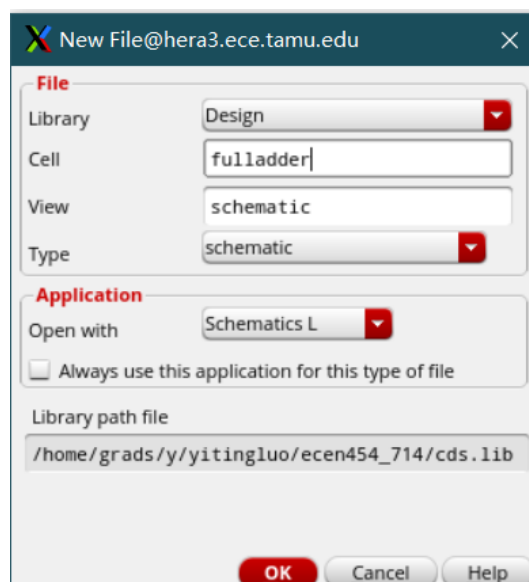


Figure 5. Creating a schematic for full-adder

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- For the adder design we will create a gate level schematic. This is called a "schematic view". In the view name field, type "schematic". Click OK on the form and a schematic view is created.
- Click on the fulladder cell in the Library Manager and you can see the schematic cellview ("fulladder schematic"). **If there is an ADE license error message “ERROR (ELI-00111) Failed to check out license 'Virtuoso_Schematic_Editor_L'.”, select “always”.**
- We can now create the first design - a gate-level schematic of 1-bit full adder. The adder will take three inputs: A, B and C (carry-in). The output will be two bits: SUM and CARRY.
- From the truth table the SUM and the CARRY functions can be generated. This can be simplified to $CARRY = AB + (A \text{ XOR } B) C$ and $SUM = (A \text{ XOR } B) \text{ XOR } C$.
- To start the Cadence schematic editor, called Virtuoso, (if it is not already open), hold the middle mouse button on the schematic cellview of fulladder in the library Manager, scroll down and choose "Open". At most of time the editor will pop up automatically when you add the new cell. This is where you design your schematic. To create an instance of a gate, choose Create → Instance from the editor menu. Component Browser (CB) and Add Instance window pops up.
- Choose library “NCSU_Digital_Parts” in CB. You will see lots of items like And, Flipflops, etc. You can choose any gate you want from this list. If you want OR gate, click “OR” in CB. You will see OR2, OR3... OR6. Choose OR2 if you want 2 inputs OR gate. If you click OR2, the Add Instance window will be automatically filled. Move your mouse to Virtuoso window and click at the point where you want to place the OR gate. One more clicks you will have one more OR gate. Press ESC if you do not want OR gate anymore.

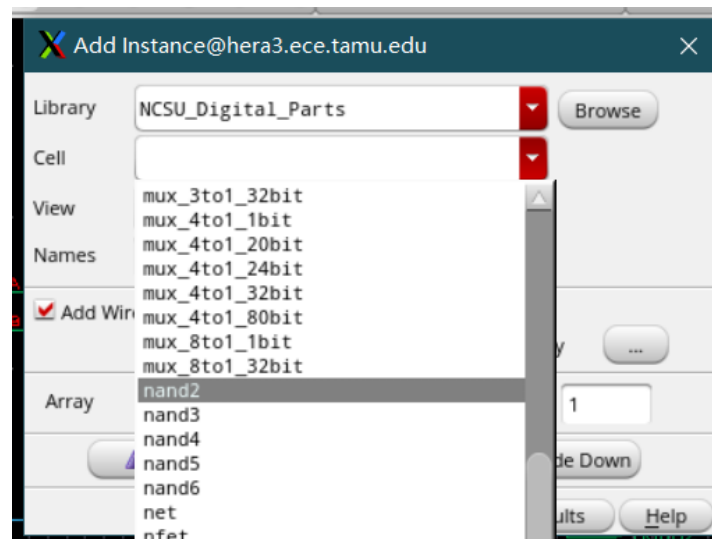


Figure 6. Add instance for full-adder schematic

NOTE: Use only XOR and NAND2 (you will have to figure out how to replace the AND and OR gates with only NAND2 gates) for the full adder. This is to maintain consistency among all students and it also means you have to implement only the XOR and NAND2 layouts. (Remember: step 2 in section 2 of Lab0 will ask you to implement layouts of all the logic gates

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you used in the schematic design. Not only will you have to design fewer layouts (NAND2 instead of AND and OR) but also the implementation of a NAND2 layout can be easier than that of an AND or OR layout as you will find in the following weeks). Also, try to minimize the number of gates used in your design. A full adder circuit can be composed of totally 5 gates of XOR and NAND2.

- Place as many instances of this gate as required. When you finish press ESC key. If you have placed some extra instances, then go to Edit > Delete in the Composer menu and click on the instance to delete. To exit from the Delete mode press ESC. To save the design click on File > Check and Save. Always save your work when you finish some important operations.
- Place instances of all the gates required and make you familiar with Add Instance and Delete Instance functions. Make sure you have plenty of space left for wiring your schematic. You can Zoom In and out to fit a part or whole of your design using View → Zoom In and Zoom Out respectively.
- We now have to wire the design. Before that we need to place input and output pins. The pins are for the input output connection of this design and also help for the simulation. We need five pins, three input pins A, B, C and two output pins SUM and CARRY. Select Create → Pin in the Virtuoso menu. Add Pin form pops up. Type “A B C” in the Pin Names field. Make sure that the Direction of the pin is “input”. Go to the schematic window and place the three input pins wherever you feel appropriate. (You might have to zoom).

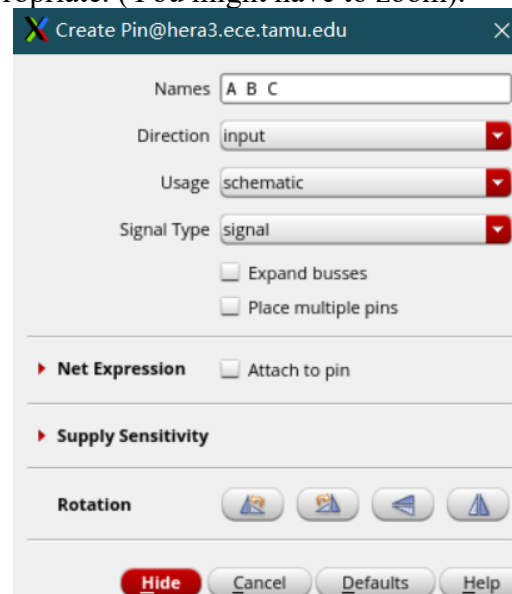


Figure 7. Add input pin for schematic.

- Note that the pins A, B and C are different from the A, B and C you see on the instances of the gates. These (the later) are for the gates themselves, one level below the hierarchy. In the same way, create the output pins “SUM and CARRY” but remember to change the direction field in the Add Pin form to “output”.
- Before wiring let’s see some Composer functions. You can move and copy the instances using Edit > Move and Edit > Copy functions. Also, you can undo the changes made using Edit > Undo. The CIW and the lower corner of the Composer will guide you.

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- Finally let's wire the schematic. Choose "Create> Wire (narrow)" in the window. Click on the A pin, a wire follows you. Take it to the point you want it to go and notice that the editor does automatic routing for you. Place it at the output of the proper gate. Similarly wire the rest of the circuit. Note that the wires can overlap without making contacts unless you explicitly do so. To make sure you wire to the exact place you want, notice that there is a diamond appearing when you wire to some place.
- You can connect up to 3 wires at one branch of wire. Pins are to be connected to instances thru wires, and if you just put the Pins on the input/output pins of a gate instance, they are not actually connected. Always use wires to make connections between instances. For example, your input is B[3:0] for the 4-bit adder, you could use bus pin option to connect your input B[3],B[2],B[1],B[0] independently like below:

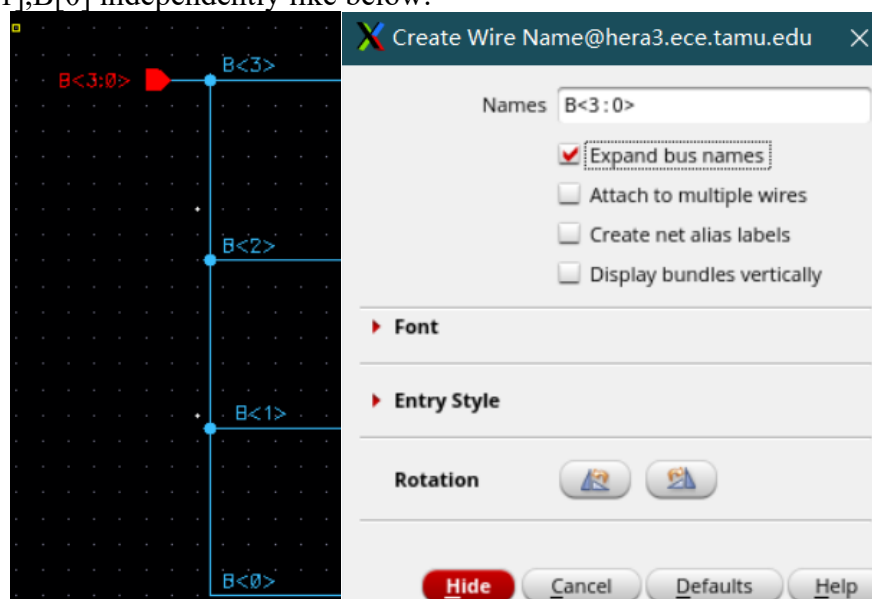


Figure 8. Add bus pin

Wire the rest of the circuit on your own. To save the design click on File > Check and Save. Look in the "CDS.log" Window for error messages. If there were errors, there would also be cross markers blinking on the schematic window. Try to correct the errors and ask your TA if you need help.

We next create a symbol view of the schematic. Choose Create > Cellview > From Cellview

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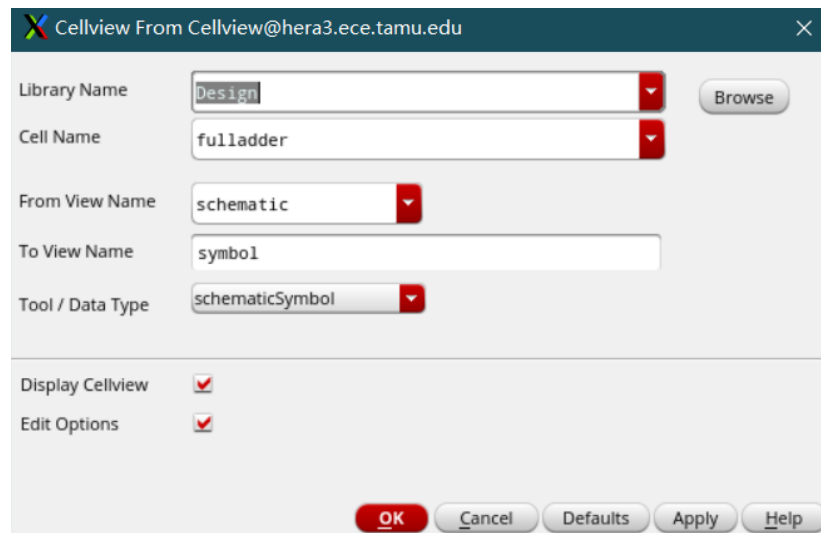


Figure 9. Creating a symbol for schematic view

This creates a symbol view of the fulladder in the Library. This can be observed in the Library manager. Observe Design > fulladder > symbol. This symbol consists of the fulladder block along with ports standing for the pins that were used in the design (in this case them being A, B, C as inputs and SUM, Carry as outputs). You could also edit it to make it look much better.

5. Lab Report

Take a plot of all the schematic designs and symbols (fulladder, 4bit adder, and 8bit adder). You may either screenshot your design or export the schematic.

Lab reports should be in PDF format and submitted to Canvas. Due date is related to your registered lab session. Late penalty is 5% per day.

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6. Appendix

Cadence tips:

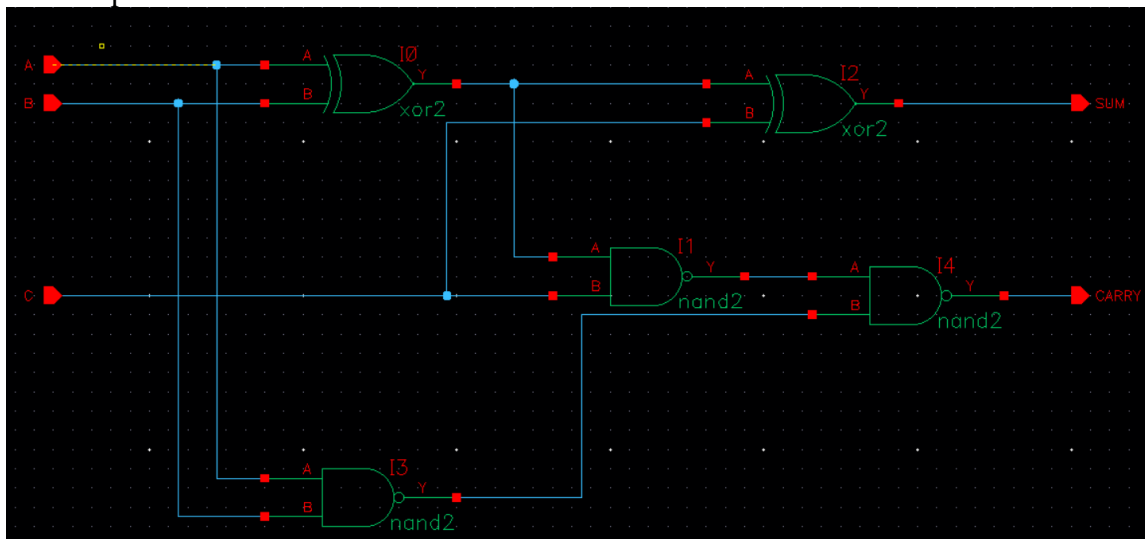
1. DO NOT KILL cadence process if the machine hangs. This generates view locks, which won't allow you to open the views in edit mode.

How to remove the lock files? Follow these steps: 1) Exit Cadence (if you have it open) 2) Go to your design folder (for example lab4 folder) and delete any files in the cell folder with the .cdslck and .oacache format. 3) Now start cadence, you would be able to edit your file.

2. Always SAVE your work. Whenever working continuously, do not wait till end to save. If cadence crashes your work will be lost.

1. Please make sure you log off using “exit” when you leave since other students cannot access the machine if you do not log off and leave.
2. If you met license error, there should be a 'cdsenv' file in your home directory. You can change the license (to always) in that file.
 - Go to home directory.
 - type in 'gedit ~/.cdsenv'
 - Change the code license *"never"* to *"always"*

Sample Schematic of Full Adder:



Grading Rubric of Lab 1 (for reference)

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1. Schematic – full adder (2); 4-bit adder (3); 8-bit adder (2)
2. Symbol – full adder (1); 4-bit adder (1); 8-bit adder (1)

If you meet the problem for “Min Width Error” or similar problem, select “Attach to a technology library”, and then change the Design library to the one you create your document, “Design” for example.

